

LEVEL 1 — Pandas Introduction

Easy Line-by-Line Notes, Code, Output & Interview Questions

1. What is Pandas?

Pandas is a Python library used for working with data.
It makes data easy to read, clean, analyze, filter, and organize.

Pandas mainly works with two objects:

1. Series — one-dimensional data.
2. DataFrame — two-dimensional table-like data.

Code:

```
import pandas as pd

data = [10, 20, 30]
s = pd.Series(data)

print(s)
```

Output:

```
Output:
0    10
1    20
2    30
dtype: int64
```

Interview Tip: Pandas is a Python library for data manipulation and analysis. It provides Series and DataFrame data structures.

2. Why use Pandas?

Real-world data is often stored in tables. Pandas helps us read CSV/Excel files, inspect data, select rows and columns, filter data, handle missing values, remove duplicates, group data, and merge datasets.

Code:

```
import pandas as pd

data = {
    "Name": ["Amit", "Neha", "Raj"],
    "Marks": [85, 92, 78]
}

df = pd.DataFrame(data)
print(df)
```

Output:

```
Output:
   Name  Marks
0  Amit     85
1  Neha     92
2   Raj     78
```

Interview Tip: Pandas reduces the amount of code needed for common data-cleaning and analysis tasks.

3. Pandas vs NumPy

NumPy is mainly designed for numerical arrays and mathematical operations. Pandas is mainly designed for labeled and tabular data. Pandas works closely with NumPy.

Code:

```
import numpy as np
import pandas as pd

a = np.array([10, 20, 30])
s = pd.Series([10, 20, 30])

print(a)
print(s)
```

Output:

```
Output:
[10 20 30]

0    10
1    20
2    30
dtype: int64
```

Interview Tip: NumPy → arrays and numerical computation. Pandas → Series/DataFrame and data analysis.

4. Installing Pandas

If Pandas is not installed, install it from Terminal, Command Prompt, or Anaconda Prompt.

Code:

```
pip install pandas
```

Output:

```
Output example:
Successfully installed pandas

Anaconda users can also use:
conda install pandas
```

Interview Tip: Do not normally write pip install inside a Python program. Run it in a terminal or notebook command cell.

5. Importing Pandas

The common import statement is:

Code:

```
import pandas as pd
```

Output:

```
No output is normally displayed because the statement only imports the library
```

Interview Tip: 'pd' is the conventional short alias for pandas.

6. Checking Pandas Version

You can check the installed Pandas version with:

Code:

```
import pandas as pd

print(pd.__version__)
```

Output:

```
Output example:
2.x.x

The exact version depends on your installation
```

Interview Tip: __version__ is useful when checking compatibility or reporting your Python environment.

7. Jupyter Notebook Setup

Basic steps:

1. Install Anaconda or Jupyter.
2. Open Jupyter Notebook.
3. Create a Python notebook.
4. Run the import statement.
5. Execute cells with Shift + Enter.

Code:

```
import pandas as pd

print("Pandas is ready!")
```

Output:

```
Output:
Pandas is ready!
```

Interview Tip: Jupyter Notebook provides an interactive environment where Python code can be executed cell by cell.

8. Creating Your First Pandas Program

Here is a simple DataFrame example.

Code:

```
import pandas as pd

data = {
    "Name": ["Amit", "Neha", "Raj"],
    "Age": [22, 21, 23]
}

df = pd.DataFrame(data)

print(df)
```

Output:

```
Output:
   Name  Age
0  Amit   22
1  Neha   21
2   Raj   23
```

Interview Tip: Line by line:

1. import pandas as pd → imports Pandas.
2. data = {...} → creates a dictionary.
3. pd.DataFrame(data) → converts the dictionary into a DataFrame.
4. df → stores the DataFrame.
5. print(df) → displays the table.
6. 0, 1, 2 → default row indexes.
7. Name and Age → column names.

LEVEL 1 — Quick Interview Revision

Q: What is Pandas?

A: A Python library used for data manipulation and analysis.

Q: What are the two main Pandas data structures?

A: Series and DataFrame.

Q: What is a Series?

A: A one-dimensional labeled data structure.

Q: What is a DataFrame?

A: A two-dimensional labeled table with rows and columns.

Q: Why is Pandas used?

A: For reading, cleaning, transforming, analyzing, and summarizing data.

Q: Pandas vs NumPy?

A: NumPy focuses on numerical arrays; Pandas focuses on labeled/tabular data.

Q: How do you import Pandas?

A: import pandas as pd

Q: How do you check Pandas version?

A: print(pd.__version__)

Q: How do you create a DataFrame?

A: pd.DataFrame(data)

Q: Why use 'pd'?

A: It is the conventional short alias for pandas.

Mini Practice Program

```
import pandas as pd

data = {
    "Name": ["Amit", "Neha", "Raj"],
    "Marks": [80, 95, 75]
}

df = pd.DataFrame(data)

print(df)
print(df["Marks"])
```

Expected result: first the complete table is displayed, then only the Marks column is displayed.